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Thought for 2m 53s >

Here is a complete English translation of the chat record in your PDF. I preserved the mathematical notation and overall structure, including the exploratory note that appears before the final answer.  Erdos#1183

Set Theory Problem Solving

User:

I am currently studying the following:

Let $f(n)$ be maximal such that in any 2-colouring of the subsets of $\{1, \dots, n\}$ there is always a monochromatic family of at least $f(n)$ sets which is closed under taking unions and intersections. Estimate $f(n)$.

Let $F(n)$ be defined similarly, except that we only require the family be closed under taking unions. Estimate $F(n)$. In particular, is it true that $F(n) \geq n^{\omega(n)}$ for some $\omega(n) \rightarrow \infty$ as $n \rightarrow \infty$, and $F(n) < (1 + o(1))^n$?

Don't search the internet. Please solve it.

Preliminary thought summary appearing in the PDF:

Let me split the problem into two parts: first, the size $f(n)$ of a sublattice closed under both unions and intersections; second, $F(n)$, where only closure under unions is required. I will first look for general lower bounds, then construct colourings to obtain upper bounds, and see whether the order of magnitude can be pinned down more clearly.

I will first focus on $F(n)$, rewriting “union-closed” in the language of posets / join-semilattices in order to capture the structure. At the same time, I will try to obtain an upper bound more concrete than merely “subexponential,” together with an explicit lower bound.

A new upper-bound direction has already appeared: if we colour by the parity of the set size, then any monochromatic union-closed family has size at most $2^{\lfloor n/2 \rfloor}$, so your guess $F(n) < (1 + o(1))^n$ looks quite possibly false. On another front, small-scale exhaustive search gives

$$f(1) = F(1) = 1, \quad f(2) = F(2) = 2, \quad f(3) = F(3) = 2, \quad f(4) = F(4) = 3,$$

so I am also probing whether they might actually equal $\lceil (n+1)/2 \rceil$.

The small-scale computation also advances to $n = 5$: one already sees that $f(5) = 3$ while $F(5) = 4$, so the two quantities begin to diverge from this point onward. Another structural line of analysis also pushes $f(n)$ down to

$$f(n) \leq Cn \log n.$$

Thought for 64m 30s

Write $\log = \log_2$. I can prove the following estimates:

$$\frac{n+1}{2} \leq f(n) \leq (4 + o(1)) n \log n,$$

and

$$\frac{n+1}{2} \leq F(n) \leq 2^{(1+o(1))(\log n)^2} = n^{(1+o(1)) \log n}.$$

Therefore, for the half you asked about,

$$F(n) < (1 + o(1))^n,$$

the answer is **yes**.

As for the other half,

$$F(n) \geq n^{\omega(n)},$$

the argument below does not yield that; at present I can only give the linear lower bound $\frac{n+1}{2}$.

1. First, two obvious lower bounds

Take a maximal chain in the Boolean lattice $2^{[n]}$:

$$\emptyset \subset \{1\} \subset \{1, 2\} \subset \cdots \subset [n].$$

This chain has $n + 1$ sets. Under any 2-colouring, one colour appears on this chain at least $\lceil (n + 1)/2 \rceil$ times.

Moreover, any subfamily of a chain is automatically closed under both unions and intersections (because the union is the larger one, and the intersection is the smaller one).

Hence

$$f(n) \geq \frac{n + 1}{2}.$$

Likewise, a chain is certainly closed under unions, so

$$F(n) \geq \frac{n + 1}{2}.$$

2. Upper bound for $f(n)$: $f(n) \leq (4 + o(1)) n \log n$

2.1 Structure of sublattices

If $L \subseteq 2^{[n]}$ is closed under unions and intersections, then it is a finite distributive lattice. Standard Birkhoff representation tells us:

- There exists a finite poset P such that $L \cong J(P)$, where $J(P)$ is the lattice of ideals of P .
- Since L is concretely embedded in a Boolean lattice, the coordinates $1, \dots, n$ split into pairwise disjoint blocks

$$B_0, B_\infty, \{B_p : p \in P\},$$

such that every set in L can be written as

$$B_0 \cup \bigcup_{p \in I} B_p, \quad I \in J(P).$$

So counting sublattices of a Boolean lattice is equivalent to:

first choosing a poset P , and then assigning the n coordinates to these blocks.

2.2 Width is controlled by size

Let $|L| = s$, i.e. $|J(P)| = s$. If P has an antichain A of size w , then for every subset $E \subseteq A$ we obtain a distinct ideal; hence

$$s \geq 2^w.$$

Therefore

$$w(P) \leq \log s.$$

In other words: if a sublattice has size s , then the width of its associated poset is at most $\log s$.

2.3 Counting posets of width at most w

Let $|P| = m$, with width $\leq w$. By Dilworth, we may decompose P into w chains.

First estimate the number of choices of an "ordered chain decomposition": at most

$$m! w^m.$$

(Choose a permutation, then assign to each element a chain; the order inside each chain is then fixed.)

Now fix two chains

$$C = (c_1 < \cdots < c_a), \quad D = (d_1 < \cdots < d_b).$$

For each c_t , the set $\{j : d_j < c_t\}$ must be an initial segment of D , and $\{j : c_t < d_j\}$ must be a final segment; hence the relations between the two chains are determined by two monotone sequences, giving at most

$$\binom{a+b}{a}^2 \leq 2^{2(a+b)}$$

possibilities.

Multiplying over all pairs of chains, and noting that

$$\sum_{i < j} (a_i + a_j) = (w-1)m,$$

we obtain: for a fixed chain decomposition, the total number of possible cross-chain relations is at most

$$2^{2(w-1)m}.$$

Thus, the number of posets of width $\leq w$ on m labelled elements is at most

$$m! w^m 2^{2(w-1)m}.$$

2.4 Now insert the coordinates

Given such a P , each coordinate $x \in [n]$ only needs to decide in which block it lies:

- B_0 ,
- B_∞ ,
- or some B_p with $p \in P$.

So there are at most

$$(m+2)^n$$

choices.

Therefore, the total number N_s of sublattices of size exactly s satisfies (writing $w = \lfloor \log s \rfloor$):

$$N_s \leq \sum_{m \leq n} (m+2)^n m! w^m 2^{2(w-1)m}.$$

Since $m \leq n$, $m! \leq n^n$, $w \leq \log s$, and $2^{2w} \leq s^2$, we get

$$N_s \leq n(n+2)^n n^n (\log s)^n s^{2n}.$$

That is,

$$\log_2 N_s \leq 2n \log s + 2n \log n + n \log \log s + O(n).$$

2.5 Random 2-colouring

Now colour $2^{[n]}$ randomly with two colours. The probability that a fixed s -element sublattice is monochromatic is

$$2^{1-s}.$$

So the expected number of monochromatic s -element sublattices is at most

$$2^{1-s} N_s.$$

The counting above shows that when

$$s \geq (4 + \varepsilon)n \log n,$$

the exponent $-s$ dominates the remaining terms, and hence

$$2^{1-s} N_s = o(1).$$

Summing over all $s \geq (4 + \varepsilon)n \log n$ still gives $o(1)$. Therefore there exists a 2-colouring with no monochromatic sublattice of size exceeding $(4 + \varepsilon)n \log n$.

Hence

$$f(n) \leq (4 + o(1)) n \log n.$$

Altogether:

$$\frac{n+1}{2} \leq f(n) \leq (4 + o(1)) n \log n.$$

3. Upper bound for $F(n)$: $F(n) \leq 2^{(1+o(1))(\log n)^2}$

Here we require only closure under unions, so the structure is much freer; however, one can use a very clean VC/Sauer argument.

3.1 A large union-closed family forces a large free join-semilattice

Write

$$B_r^- := \{\emptyset \neq X \subseteq [r]\}.$$

Equipped with union, this is the free join-semilattice on r generators (with the empty set removed).

Key lemma. If a union-closed family $U \subseteq 2^{[n]}$ satisfies

$$|U| > \sum_{i=0}^{r-1} \binom{n}{i},$$

then U contains a copy of B_r^- .

Proof. By Sauer-Shelah, U shatters some r -element set $T \subseteq [n]$. Therefore, for each $i \in T$, we may choose $A_i \in U$ such that

$$A_i \cap T = \{i\}.$$

For each nonempty $X \subseteq T$, define

$$U_X := \bigcup_{i \in X} A_i.$$

Since U is union-closed, we have $U_X \in U$. Also,

$$U_X \cap T = X,$$

so these U_X are all distinct, and satisfy

$$U_X \cup U_Y = U_{X \cup Y}.$$

This is exactly a copy of B_r^- . ■

So, as long as a colour class contains no monochromatic B_r^- , we automatically get

$$F(n) \leq \sum_{i=0}^{r-1} \binom{n}{i}. \quad (1)$$

3.2 Use random colouring to eliminate large B_r^-

We now estimate how many copies of B_r^- there are in $2^{[n]}$.

A copy of B_r^- is determined by r generators $G_1, \dots, G_r$. For each ground element $x \in [n]$, we only need to record which generators contain it; this gives a subset

$$I_x \subseteq [r].$$

So there are at most

$$(2^r)^n = 2^{rn}$$

choices. In other words, the number of copies of B_r^- is at most 2^{rn} .

A fixed B_r^- has $2^r - 1$ points, and the probability that it is monochromatic is

$$2 \cdot 2^{-(2^r-1)} = 2^{2-2^r}.$$

Therefore, under a random 2-colouring, the expected number of monochromatic copies of B_r^- is at most

$$2^{rn+2-2^r}.$$

As soon as

$$2^r \gg rn,$$

this expectation tends to 0.

Take

$$r = \lceil \log n + 2 \log \log n \rceil.$$

Then $2^r \asymp n(\log n)^2$, and hence $2^r \gg rn$. So there exists a colouring with no monochromatic B_r^- .

Substituting into (1),

$$F(n) \leq \sum_{i < r} \binom{n}{i}.$$

Using the standard estimate

$$\sum_{i < r} \binom{n}{i} \leq \left(\frac{en}{r}\right)^r,$$

we obtain

$$F(n) \leq 2^{(1+o(1))(\log n)^2} = n^{(1+o(1)) \log n}.$$

In particular,

$$F(n) < (1 + o(1))^n.$$

4. Final summary

So what I obtain is:

$$\frac{n+1}{2} \leq f(n) \leq (4 + o(1)) n \log n$$

and

$$\frac{n+1}{2} \leq F(n) \leq 2^{(1+o(1))(\log n)^2}.$$

Therefore:

- $f(n)$ is at least linear and at most $n \log n$;
- $F(n)$ is at least linear and at most quasi-polynomial;
- for your question $F(n) < (1 + o(1))^n$, the answer is yes;
- for your question $F(n) \geq n^{\omega(n)}$, I still cannot prove it; with the present method I only get a linear lower bound.

If you would like a more “research-note style” version, the next most natural thing would be to fully expand the representation of the “sublattice = weighted ideal lattice” part for $f(n)$ into an independent chain of lemmas.

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来源

